

**(Automated) Assessment of the Narrative Description of Analysis Code –
A CodeBot Validation Study**

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Preregistration of Bachelor's Thesis

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Introduction

In the light of psychology's very own replication crisis, many researchers find themselves between a rock and a hard place trying to find ways to move towards a reliable and robust future. While estimated replicability rates across the psychological literature are noteworthily low (see e.g. Open Science Collaboration, 2015), categorically deeming all literature that has come before as untrustworthy would be hastily pulling the rug from under one's feet. Instead, employing skills and tools that help assess the trustworthiness of past and future research could foster an efficient culture of transparency and openness assessment in the scientific community. RegCheck and its sibling CodeBot – both developed by Cummins et al. (see Cummins et al., 2026 for an introduction to RegCheck for pre-registration/paper comparison) – aim to offer an accessible way for researchers to perform such fundamental trustworthiness assessments. CodeBot uses natural language processing and large language model (LLM) to assess the degree to which analyses described in the written manuscript match analyses in the corresponding analysis code. The aim of this study is to provide first evidence for the validity of CodeBot's assessment by comparing its judgement to human coding of code-paper consistency. In this pre-registration, I will first outline the general sample characteristics and coding rationale, then I will outline the results of a pilot run from two coded papers, followed by a determination of the sample for the main study, an a posteriori precision analysis and lastly the research questions for the main study as well as the statistical analysis plan.

Method

Disclaimer: As of now, the data from the human coder has already been collected. The corresponding file can be found under “./data/raw/main/human/” in the OSF repository archive. Analyses by CodeBot and the automated comparisons have yet to take place. No analyses have been performed on the final sample.

Sample Characteristics

Inclusion criteria

The samples for both the piloting and main study contain papers fulfilling the following criteria: (a) published in a psychological journal (b) with a Transparency and Openness Promotion (TOP) factor of at least two in the domain “Analysis Code Transparency”, with (c) analysis code (claimed to be) openly available, (d) in the coding language R and (e) a publishing date no more recent than either the December issue of the journal and/or the 16th of December 2025.

Exclusion criteria

To ensure that the scale of this work was within scope for a bachelor’s thesis, papers were excluded when they did not contain one of the following statistical methods as their main analysis: t-tests, z-tests, univariate (repeated measures) analyses of variance (ANOVAs), univariate linear regression, Chi²-tests (tests of proportion, not of model fit), univariate logistic regression, exploratory factor analysis, equivalence tests, Wilcoxon-rank-sum-test, Wilcoxon-signed-rank-test or Kruskal-Wallis-test.

Papers of type review, meta-analysis, tutorial, theoretical piece, editorial, obituary, comment/response and call for papers were excluded. Power analyses and manipulation-, attention- or sanity checks of the data were not regarded as main analyses and therefore were not coded. If any of the tests (e.g. a t-test) were “nested” inside otherwise excluded analyses, for example in the form of regression weight significance testing, which is usually carried out in a single R function call, they were not included either. If the information provided insufficiently described a procedure, the corresponding model was also not coded.

Coding Process

Papers are read and searched for information on the statistical analyses. Each statistical analysis is assigned a unique ID and coded on the following four dimensions: model specification, variable specification, transformation and inference criteria.

Model specification refers to the type of statistical model/analysis analogue to the list from the inclusion criteria. Each variable that is contained in the statistical model gets assigned its own row for variable specification. Processing of data prior to analysis (e.g. calculating a sum score across multiple items) which is explicitly described as part of the measurement/analysis section and not as preliminary wrangling, is coded as a transformation. Lastly, inference criteria, i.e. explicitly stated (alpha) significance levels and confidence intervals were coded. For each claim, a quote and reference from both the written manuscript as well as the analysis script is supplied, together with information on whether data is needed for the assessment. The final judgement is recorded on the target variable *results* (1 = *paper-code match*, 0 = *paper-code mismatch*, NA = *indeterminable*). Whenever analyses are mentioned multiple times, only the first occurrence is coded but additional, new information is included in the manuscript quotes. Analyses are recorded with highest granularity. For example, when a zero-order correlation table is presented, each correlation is considered a separate model.

Note, that the aim of this study is not the examination of computational reproducibility which therefore does not constitute a mismatch. Moreover, if there are additional details in the code that are not described in the manuscript, this is not considered a mismatch since the assessment is unidirectional, i.e. strictly going from the narratively reported analyses in the manuscript to the code and not the other way around.

Both raters (human/CodeBot) perform the coding blinded to the other's judgement, resulting in two separate .csv files. Then, to enable swift comparisons between the two files,

an adjudication program is used which programmatically matches and compares the best fitting observations across the .csv files, making judgements on whether there is agreement between the raters. The program's output is a .csv file holding information on the different observations from both raters as well as an additional variable assessing the two coders' agreement on each rating (1 = *match*, 0 = *mismatch*). For further details on the employed programs, see the "methods" folder in the OSF repository archive.

Piloting and A Priori Precision Analysis

To estimate the achievable precision of said agreement, pilot coding and analysis of two papers was performed (see `./data/raw/pilot/` in the OSF repository archive). Across these two papers, the mean number of coded analyses was 38.5 with a standard deviation of 10.5. A generalized linear mixed model (GLMM) with a logit link function was fit to the pilot data to deliver a first estimand of the expectable agreement between the raters' judgements. The fitted model estimates the expected probability of agreement to be 98.3% (95% CI [95.5%,99.4%]), in cases where both human and CodeBot ratings are present. (See `./code/pilot/analysis.qmd` in the OSF repository archive).

Sample Determination

Searching the TOP database (<https://www.topfactor.org/journals>) returned 23 journals when looking for "psycho" in the Analysis Code Transparency domain. One was a duplicate, leaving 22 unique journals. Another 5 were excluded, namely: *American Psychologist*, *Frontiers in Psychology*, *Personality and Social Psychology Review*, *Psychological Bulletin* and *Psychological Methods*.

This left 17 journals: *Meta-Psychology*, *Advances in Methods and Practices in Psychological Science*, *Applied Psycholinguistics*, *Clinical Psychological Science*, *Collabra: Psychology*, *Comprehensive Results in Social Psychology*, *Experimental Psychology*, *Global Environmental Psychology*, *Health Psychology*, *Journal of Experimental Psychology*:

Applied, Journal of Experimental Psychology: General, Journal of Experimental Psychology: Human Perception and Performance, Journal of Personality and Social Psychology, Personality and Social Psychology Bulletin, Psychology and Aging, Psychology, Public Policy and Law, Social Psychology. The first paper from each journal that met all the criteria and was published by the 16th of December 2025 was coded. To accommodate the time constraints of a bachelor's thesis, the final sample size for this project was set to the number of eligible journals ($N = 17$). Table 1 holds the list of all coded papers.

Table 1

Abbreviated Journal Titles and Corresponding DOI of Sampled Papers

Journal	Authors	DOI
<i>MP</i>	Hussey, I.	https://doi.org/10.15626/MP.2023.4008
<i>AMPPS</i>	Bloy et al.	https://doi.org/10.1177/25152459251378420
<i>APL</i>	Yang et al.	https://doi.org/10.1017/S0142716425100313
<i>CPS</i>	Yeung et al.	https://doi.org/10.1177/21677026251315118
<i>CPSY</i>	Radtke et al.	https://doi.org/10.1525/collabra.151266
<i>CRSP</i>	Evans et al.	https://doi.org/10.1080/23743603.2025.2507434
<i>EP</i>	Besken & Filiz	https://doi.org/10.1027/1618-3169/a000646
<i>GEP</i>	Berger et al.	https://doi.org/10.5964/gep.10895
<i>HP</i>	Cha et al.	https://doi.org/10.1037/hea0001524
<i>JEPA</i>	Carvalho & Goldwin	https://doi.org/10.1037/xap0000523
<i>JEPG</i>	Akan et al.	https://doi.org/10.1037/xge0001836
<i>JEPHPP</i>	Münster & Frings	https://doi.org/10.1037/xhp0001359
<i>JPSP</i>	Nguyen & Ames	https://doi.org/10.1037/pspa0000456
<i>PSPB</i>	Arvilles et al.	https://doi.org/10.1177/01461672241257139
<i>PA</i>	Hsieh et al.	https://doi.org/10.1037/pag0000920
<i>PPPL</i>	Helm & Spearing	https://doi.org/10.1037/law0000456
<i>SP</i>	Mancassola et al.	https://doi.org/10.1027/1864-9335/a000580

A Posteriori Precision Analysis

To estimate the precision that could possibly be achieved with this sample size, the previously mentioned sample characteristics (M and SD of analyses per paper, agreement estimand from the piloting papers) were used to simulate 2000 sets of agreement data. Then, a GLMM with a logit link function was fit to each dataset (Wilkinson notation: $\text{agree} \sim 1 + (1|\text{paper_id})$) and the quantile based 95% confidence interval of the probability intercept was calculated. An arbitrarily chosen threshold of a CI width smaller than or equal to 0.05 was chosen to indicate sufficiently precise estimation. Since all CIs were narrower than 0.035, 17 papers will still yield a precise estimation under the assumption that the 98% observed agreement rate during piloting is an appropriate estimand of the true agreement rate and not a mere product of chance (See “./code/precision_analysis/bachelor_sim.qmd and “./data/raw/precision_analysis/” in the OSF repository archive).

Research Questions

Q1: How often does the narrative description of statistical analyses in the manuscript match the analyses reported in the analytic scripts according to the human and CodeBot rating, respectively? → Descriptive examination of (mis-)matches for human and CodeBot ratings.

Q2: How high is the estimated agreement of the human and CodeBot ratings of the judgement of (mis)matches between the narrative description of statistical main analyses and the respective analytic code? → GLMM estimating agreement while considering nesting effects per paper.

Q3: What could explain why CodeBot coded analyses the human did not code and vice versa? → Qualitative examination of 3% randomly sampled specifications that were code by CodeBot or the human coding only.

Q4: (Why) are there reoccurring characteristics of the analyses that lead to disagreement in ratings? → Quantification of which categories show the highest rates of disagreement an examination of 10% randomly selected observations with disagreement.

Statistical analysis

Outcome variable

The variable of interest is agreement between the two coders. This will be represented by a binary variable (1 = *match*, 0 = *mismatch*). Since there is only one human rater involved, there is no “ground truth” for the data. Note that the “pairing” of human/CodeBot coding is automated because the time constraints of this thesis do not allow for manual matching of the respective ratings.

Data cleaning and processing

Data wrangling and analysis will be conducted in R using RStudio (Posit Team, 2026). Rows which are empty, meaning all empty strings or only NAs on all variables, will be removed. If there is reason to believe that data quality is compromised due to human error introduced by manual coding and the fact that only one person was coding the data, the respective observations get flagged. When the error is minor, e.g. a mismatched ID or dimension, it is flagged but corrected. In these cases, a written explanation (likely as part of supplementary materials) is provided to ensure transparency. If necessary, data will be converted into tidy format (Wickham, 2014).

Descriptive analysis

To get an overview of the data, the following descriptive statistics will be calculated: (a) number of observations coded by CodeBot/human respectively, (b) number of observations coded by both coders, (c) agreement rate for judgements (match/mismatch/indeterminable) made by both raters.

Research Question 3 and 4 will mostly be assessed qualitatively by re-examining the corresponding sections in the manuscript and looking for repeating characteristics of the data that as well as exploring and verifying its contents.

Inferential analysis

Only instances where human and CodeBot assessment is available will be considered for the GLMM. The GLMM will be specified as follows, using the lme4 (Bates et al. 2015) package: `glmer(agree ~ 1 + (1|paper_id), data = ., family = binomial)`.

References

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